Vuthtyra (Teera) Yong

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EDUCATION

Syracuse University, School of Information Studies Syracuse, NY
M.S. in Applied Data Science, GPA: 4.0/4.0 Dec 2025
Relevant Coursework: Applied Machine Learning, Natural Language Processing, Deep Learning, Advanced Big Data Management, Big Data Analytics, Data Warehousing, Cloud Management, Business Analytics

TECHNICAL SKILLS

Languages: Python, SQL, R, C++
Machine Learning: PyTorch, TensorFlow, Hugging Face, LangChain, RAG, supervised/unsupervised learning, time series forecasting
Data & Big Data: Apache Spark, Kafka, MongoDB, Redis, Cassandra, Hadoop, Neo4j, Elasticsearch, dbt
Cloud & Dev Tools: Azure, AWS, Snowflake, Docker, Kubernetes, Git, Linux
Visualization & Web: Tableau, Power BI, Excel, Flask, FastAPI, Streamlit, React

PROJECT EXPERIENCE

ScholarAI | React, FastAPI, Azure Blob Storage, Multi-Agent Orchestration

- Built a full-stack research assistant that searches arXiv, summarizes papers, generates literature reviews, formats citations, and manages saved paper collections.
- Designed a modular agent architecture with remote summarization and shared tool integration to improve maintainability and extensibility.
- Implemented persistent save/load workflows with Azure Blob Storage and added graceful handling of external API failures to improve reliability.

Multimodal Product Search & Retrieval | Python, SigLIP, PyTorch, FAISS

- Fine-tuned a SigLIP dual-encoder on a curated subset of the Amazon Berkeley Objects dataset (11K+ products) to align image and text embeddings for semantic search.
- Built an end-to-end retrieval pipeline with FAISS indexes for text-to-image and image-to-text search, achieving **Recall@3 = 0.95** and **MRR = 0.86** on the test set.
- Analyzed learned embeddings with t-SNE visualizations and qualitative search examples to validate stronger cross-modal alignment after fine-tuning.

Amazon Product Reviews Analysis | Python, BERT, UMAP, PCA, K-Means

- Transformed 153K Amazon reviews into 768-dimensional BERT embeddings and applied dimensionality reduction to visualize sentiment and product patterns.
- Clustered review embeddings to identify major themes and used GPT-4 generated summaries to label 90 clusters and surface actionable product insights.

CERTIFICATIONS & BADGES

Microsoft Certified: Azure AI Engineer Associate (AI-102) Mar 2026
AWS Certified Cloud Practitioner (CLF-C02) Feb 2025
Microsoft Certified: Azure Fundamentals (AZ-900) Feb 2025
Snowflake Hands-on Essentials: Data Warehouse Jan 2025